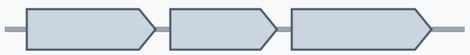
phu jack: selecting contigs by iterative search from seed proteins

A Seed proteins and gene prediction

Seed markers (protein FASTA, one or more sequences)

S1 S2 used when a trusted marker protein exists but no profile HMM does

Contigs (FASTA)

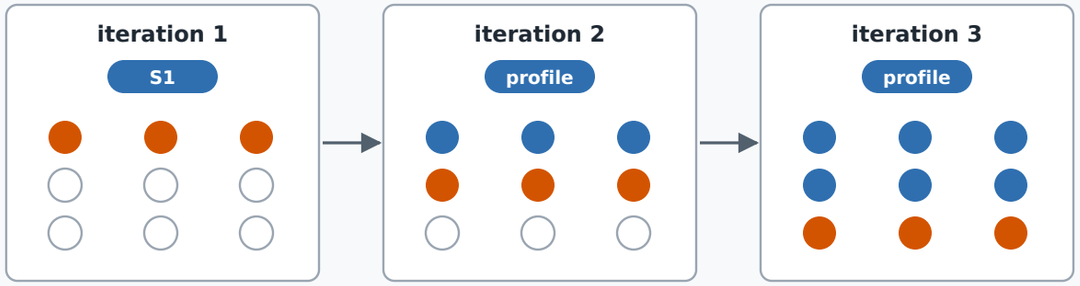


pyrodigal gene prediction
meta mode by default; --mode, --ttable

Predicted proteins (≥ 30 aa) p1 p2 p3 p4 p5

The prediction cache is shared with phu screen and phu avger

B Iterative search (PyHMMER jackhmmer), one seed at a time



newly included hit (E-value ≤ 1e-3, --inc-evalue) already in the profile

After each round the included hits are aligned into a profile, which is searched again. The search runs for up to 5 rounds (--iterations) and so reaches more distant relatives than the seed alone. Each seed is searched independently.

C Final hits and contig selection

Final hits: E-value ≤ 1e-5
--max-evalue

Best hit per contig by bit score
--top-per-contig 1

With several seeds, --combine-mode decides which contigs are kept

	S1	S2	any (default)	all	threshold ≥ 2
c1	●	●	kept	kept	kept
c2	●	○	kept	dropped	dropped
c3	○	○	dropped	dropped	dropped
c4	○	●	kept	dropped	dropped

any at least one seed's search hits the contig
all every seed's search hits the contig
threshold at least --min-seed-hits seeds hit the contig

matched contigs

D Optional: keep the final profile

The profile of the last round can be saved and used as a query elsewhere

S1 → **jackhmmer rounds** (panel B) → **final-iteration.hmm**

--save-hmm

phu screen -i other_contigs.fna final-iteration.hmm
the saved profile screens other assemblies directly, without repeating the iterative search

With one seed the file is final-iteration.hmm; with several, one profile per seed is written under final-iteration-hmms/.

jackhmmer-iterations.tsv records how many hits each round added, which shows whether the search converged or kept growing.

Outputs

matched-contigs.fna	matched-contig-ids.txt	jackhmmer-hits.tsv	jackhmmer-iterations.tsv	final-iteration.hmm
contigs that passed	their names	final hits and filters applied	summary of each round	optional

Hits gained over the rounds are sequence-similarity evidence, not a validated functional or taxonomic assignment.

Contigs, proteins and hits shown are illustrative